Exact equation

Consider the first-order equation

$$M(x,y)+N(x,y)\cdot y'=0.$$

If there is a C^2-function F such that

$$\frac{\partial F}{\partial x}=M \quad \text{and} \quad \frac{\partial F}{\partial y}=N$$

then, for a solution $y=\varphi(x)$,

$$M(x,y)+N(x,y)\cdot y'=\frac{\partial F}{\partial x}(x,\varphi(x))+\frac{\partial F}{\partial y}(x,\varphi(x))\cdot\varphi'(x)$$

$$=\frac{d}{dx}\cdot F(x,\varphi(x))=0.$$

So, we obtain the implicitly relation

$$F(x,\varphi(x))=C.$$

Moreover, by the Fubini theorem, if $M_y=N_x$, we can be guaranteed the existence of such a F. It is necessary·surficient condition.

Further, although $M_y\neq N_x$, if there is a function $\mu(x,y)$ such that

$$\frac{\partial}{\partial y}\left[\mu(x,y)\cdot M(x,y)\right]=\frac{\partial}{\partial x}\left[\mu(x,y)\cdot N(x,y)\right]$$

$$\Leftrightarrow \mu_y\cdot M+\mu\cdot M_y=\mu_x\cdot N+\mu N_x$$

$$\Leftrightarrow \mu_x\cdot N-\mu_y\cdot M=(M_y-N_x)\cdot\mu. \quad \cdots (*)$$

So, $(*)$ is necessary·surficient condition for such μ exists.
In general, the condition $(*)$ is too hard to solve, (it is PDE)
But, if $(M_y-N_x)/N$ is a function dependent to only the variable x, then the condition $(*)$ implies

$$\frac{1}{\mu}d\mu=\frac{M_y-N_x}{N}dx \Rightarrow \mu=\exp\left(\int\frac{M_y-N_x}{N}dx\right.$$

Problem.

Consider

$$\left(x \cdot e^{\frac{y}{x}} + 2 \cdot x^2 y\right) dy - y \cdot e^{\frac{y}{x}} dx = 0.$$